x86 architecture

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The x86 family

It consists of a number of processors, manufactured by Intel and others (IBM, NEC, AMD, TI, STM, Fujitsu, OKI, Siemens, Cyrix, Intersil, C & T, NexGen, UMC).

The 8086 processor was the first of the x86 family.

Within this family, *backward compatibility* is guaranteed.

This means that each processor can run any code produced for the previous processors.

x86 architecture

Software compatibility implies that x86 systems share the same logical architecture, i.e., the view of the processor from the programmer point of view (also called *Instruction Set Architecture*).

The x86 architecture can sometimes be divided into 3 types

- **x86-16: 16-bit architecture, typical of 8086**
- **x86-32: 32-bit architecture, typical of processors starting from 80386**
- **x86-64: 64-bit architecture, typical of processors starting with AMD K8.**

3

General characteristics of 8086

t was released on the market in 1978 as the evolution of 8080 and 8085 processors.

It has

- **an address space of 2^{20} = 1Mbyte**
- **16 out of 20 address pins also act as data pins**

It has a 16-bit parallelism for

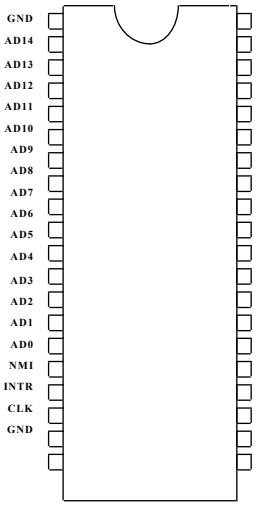
- **ALU**
- **Registers**
- **Data bus.**

4

8086 pins



Modo Max (Modo Min)



5

5

S_0, S_1, S_2

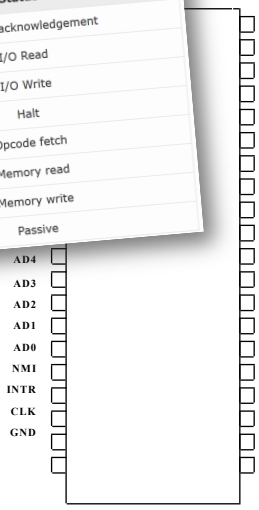
These are the status signals that provide the status of operation, which is used by the Bus Controller 8288 to generate memory & I/O control signals. These are available at pin 26, 27, and 28. Following is the table showing their status –

S_2	S_1	S_0	Status
0	0	0	Interrupt acknowledgement
0	0	1	I/O Read
0	1	0	I/O Write
0	1	1	Halt
1	0	0	Opcode fetch
1	0	1	Memory read
1	1	0	Memory write
1	1	1	Passive

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Modo Max (Modo Min)

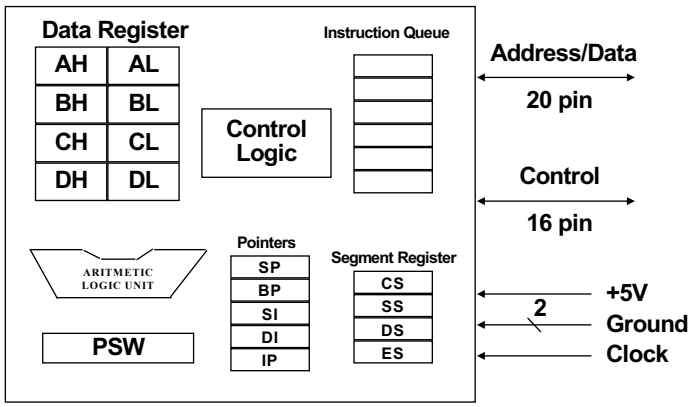




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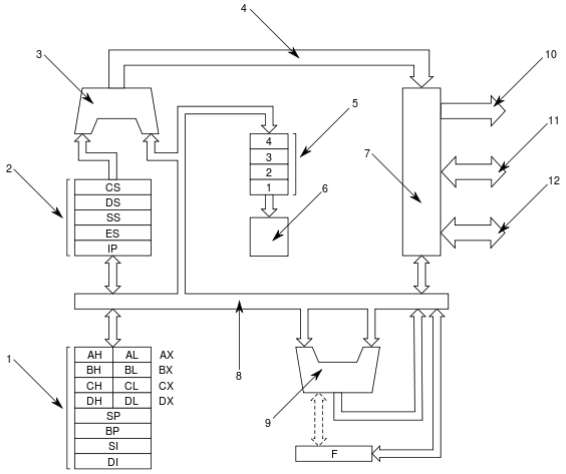
Architecture



7

7

Simplified 8088



1=main registers; 2=segment registers and IP; 3=address adder; 4=internal address bus; 5=instruction queue; 6=control unit (very simplified!); 7=bus interface; 8=internal databus; 9=ALU; 10/11/12=external address/data/control bus.

8

8

Registers

They can be divided into 3 groups:

- **data registers**
- **Index registers**
- **segment registers.**

9

Data registers

They are

- **AX (Accumulator Register)**
- **BX (Base Register)**
- **CX (Count Register)**
- **DX (Data Register).**

They are used to store operations and the result of operations.

They can be used as 16-bit registers or as a pair of 8-bit register.

BX can also be used in address calculation.

CX is also used as a counter for some instructions.

DX contains the I / O address in some I / O instructions.

10

Index registers

They are IP, SP, BP, SI and DI:

- **IP (Instruction Pointer): it contains the pointer to the first instruction to execute. IP can not appear explicitly as an instruction.**
- **SP (Stack Pointer): it contains the pointer to the top of the stack.**
- **BP (Base Pointer): it is used as the basis for accessing the stack.**
- **SI (Source Index) and DI (Destination Index) are used as index registers.**

Segment registers

They are CS, DS, ES and SS.

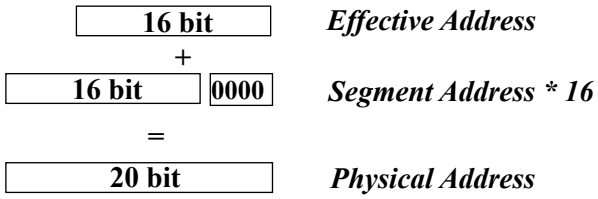
They are used to compute physical addresses to access memory.

They store pointers to the beginning of the code, data, extra data, and stack segments, respectively.

Address computation

Each time the 8086 has to generate an address to access the memory (*physical address*), it performs the following operations

- **it multiplies the contents of a segment register by 16 (shift by 4 positions)**
- **it adds the content of an index register or BX (storing the *effective address* or *offset*) and the content of a segment register.**



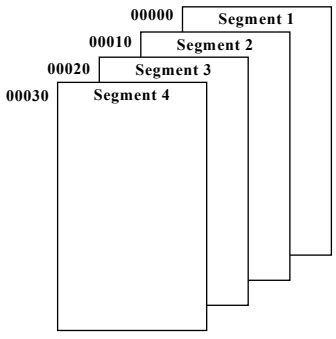
13

Segments

Memory can be considered as organized in *segments*, each of a size of 64 Kbytes.

All segments begin at an address which is a multiple of 16.

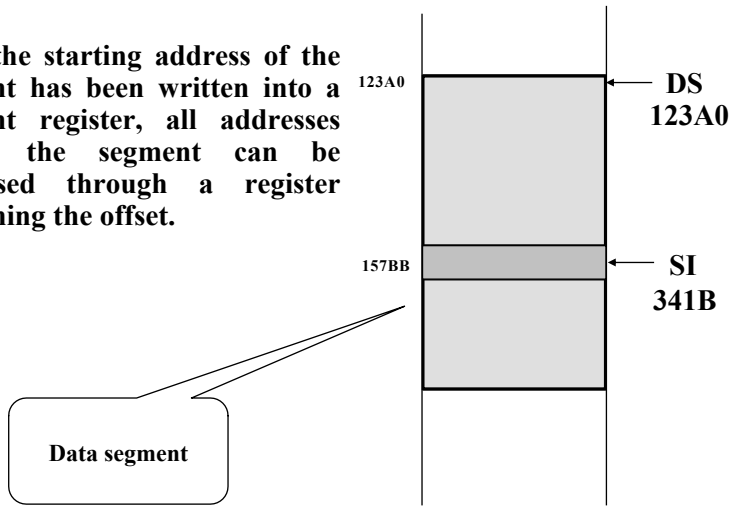
Segments can partly overlap.



14

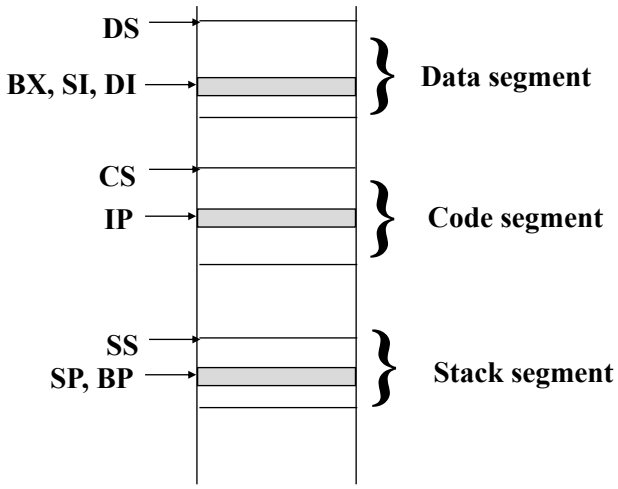
Use of Segment Registers

Once the starting address of the segment has been written into a segment register, all addresses within the segment can be expressed through a register containing the offset.



15

Memory segmentation



16

Advantages of segmentation

- Address space equal to 2^{20}, but 16-bit addresses
- Separation between data, code and stack
- Possibility to have multiple segments of the same type (data, code or stack)
- Possibility of segment overlapping, minimizing wasted memory
- Relocatability.

Process Status Word (PSW)

It is composed of 16 bits, but only 9 of them are used.

Each bit corresponds to a flag.

Flags are divided into:

- condition flags
- control flags.

				OF	DF	IF	TF	SF	ZF		AF		PF		CF

Condition flags

They are automatically written at the end of various operations:

- **SF (Sign Flag): coincides with the MSB of the result after an arithmetic instruction**
- **ZF (Zero Flag): it is 1 if the result is null, 0 otherwise**
- **PF (Parity Flag): it is 1 if the number of 1 in the 8 less significant bits of the result is even, 0 otherwise**
- **CF (Carry Flag): after any arithmetical instruction it is 1 if a carry (sum) or borrow (subtraction) has been produced; other instructions can make a particular use of CF**
- **AF (Auxiliary Carry Flag): used in BCD arithmetic**
- **OF (Overflow Flag): it is 1 if the last instruction produced an overflow.**

Control flags

They can be written and manipulated by specific instructions, and serve to regulate the operation of certain processor functions:

- **DF (Direction Flag): used by strings handling instructions;**
- **IF (Interrupt Flag): if it is 1, the maskable interrupt request is detected by the CPU, otherwise it is ignored**
- **TF (Trap Flag): if it is 1, a trap is executed at the end of each statement.**

Memory access

The 8086/8088 has an address space of 1 Mbyte.

The 8086 can access in one cycle to a byte or memory word, provided that this starts at an even address.

Accessing words aligned to an odd address requires two memory cycles.

8088 requires one cycle for each byte that is accessed in memory.

Memory access (II)

Physically, the 8086 memory is organized as two 512Kbyte banks:

- **odd bank (D15-D8)**
- **even bank (D7-D0).**

The two banks are addressed in parallel by the address lines A19-A1.

Even address data are transferred to pins D7-D0 while odd address data are transferred to pins D15-D8.

Memory access (III)

The processor generates two control signals (BHE and A0) to manage memory accesses.

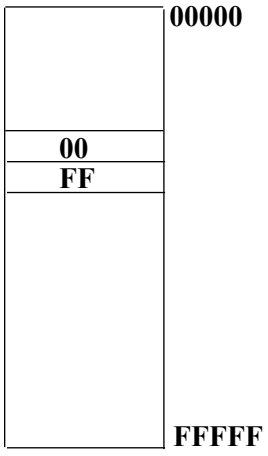
BHE	*A0*	*Memory operation*
0	**0**	**Word**
0	**1**	**Most significant byte (odd address)**
1	**0**	**Least significant byte (even address)**
1	**1**	**Forbidden**

23

Word representation

It is assumed that the data stored in a word have the least significant byte stored in the byte with lower address (*little endian* representation).

The example shows how the value FF00 is stored.



24

Reserved parts of the memory

Some parts of the memory are not free, but dedicated, or reserved.

For example, memory from 00000H to 003FFH is reserved, as it contains the Interrupt Vector Table.

Addresses from FFFF0H to FFFFBH are used to contain a jump instruction to the bootstrap program routine.

Locations from FFFFCH to FFFFFH are reserved for future use.



00000
003FF
FFFF0
FFFFC

25

Instruction format

It is very irregular.

Each instruction can occupy from 1 to 6 bytes.

Examples

NOP, STC, POP AX	1 bytes
MOV AX, BX	2 bytes
ADD AX, BX	2 bytes
JNE lab	2 bytes
CALL P1	3 bytes
MOV var, AX	3 bytes
MOV vet[DI], CX	4 bytes
JMP label (far)	5 bytes
MOV vet[SI], 10000	6 bytes

26

Timings

Instruction	Description	Clock Cycles	Number of Bytes
NEG	Reg	3	2
	Mem	16 + EA	2-4
DEC, INC	16-bit Reg	2	1
	8-bit Reg	3	2
	Mem	15 + EA	2-4
ADD, SUB	Reg → Reg	3	2

Instruction	Description	Clock Cycles	Number of Bytes
CBW	Convert byte to word	2	1
CWD	Convert word to double word	5	1
MUL	8-bit Reg	70 – 77	2
	16-bit Reg	118 – 133	2
	8-bit Mem	(76 – 83) + EA	2-4
	16-bit Mem	(124 – 139) + EA	2-4
IMUL	8-bit Reg	80-98	2
	16-bit Reg	128 – 154	2
	8-bit Mem	(86 – 104) + EA	2-4
	16-bit Mem	(134 – 160) + EA	2-4
DIV	8-bit Reg	80 - 90	2
	16-bit Reg	144 - 162	2

27

27

Timings

EA	Number of Clock Cycles
Direct	6
Register indirect	5
Register relative	9
Based indexed	
(BP)+(DI) or (BX)+(SI)	7
(BP)+(SI) or (BX)+(DI)	8
Based indexed relative	
(BP)+(DI)+DISP or (BX)+(SI)+DISP	11
(BP)+(SI)+DISP or (BX)+(DI)+DISP	12

28

28